

V2X-Sim — Sprint 5 Final Report

Collision Realism, Vehicle-to-Vehicle Communication, and a Mixed-Traffic Penetration Study

Cooperative Perception at Smart Intersections — a CARLA study of 5G NR-V2X with ETSI CPM
Release 2

CMP794 Final Project | Taha Yasin Er & Doruk Topçu
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Executive summary

Sprint 5 turned the cooperative-perception ablation from a metric with inflated, hard-to-trust numbers into a clean, reproducible safety experiment, and added vehicle-to-vehicle (V2V) communication on top of the existing roadside-to-vehicle (V2I) stack. Four correctness fixes (collision counting, driver-role behaviour, spawn overlap, hostility scope) plus the V2V layer support a full penetration study on the in-distribution map (Town05) at $p = 0.0 / 0.5 / 0.9 / 1.0$, with every cell run both with and without V2V to isolate its effect.

Two results stand out. (1) **Vehicle penetration is the dominant safety lever and it generalises:** replacing hostile human drivers with careful connected-autonomous vehicles drives collisions down monotonically (Town05 with V2V: 27.5 → 15.5 → 5.6 → 3.2), and the same trend holds on the held-out Town10 map. (2) **V2V's marginal value over roadside-only perception depends on scenario difficulty:** roughly neutral on the easy map at mid penetration, but a clear collision reduction on the harder held-out map, at the cost of a 3-7x increase in braking. The test suite grew 215 → 241, all passing.

Scope honesty. The full Town10 cross-map and all of Town01 are absent by design: at full V2V penetration the single-threaded message-fusion cost grows super-linearly and those cells cannot complete in feasible wall-time at the study's fidelity. Section 6 explains this with profiling evidence and the deliberate decision not to lower fidelity for a subset (which would break comparability).

1. Scenario and project context

The project studies **cooperative perception at signalised intersections**: roadside units (RSUs) with cameras detect road users and broadcast them to connected vehicles, which fuse those messages with their own sensing to react earlier in mixed traffic. The whole pipeline is modelled end-to-end on CARLA 0.9.16 rather than assuming an idealised channel.

Pipeline: RSU camera -> YOLO26 detector -> ETSI CPM message -> V2X channel (packet loss + latency + channel-busy ratio) -> connected-vehicle late-fusion -> braking decision.

Roles:

- **HDV** — human-driven vehicle (CARLA Traffic Manager + driver profiles). The conflict source.
- **CAV** — connected autonomous vehicle: runs cooperative-perception fusion + braking, and (Sprint 5) the V2V layer.
- **RSU** — one overhead camera per intersection running the detector and broadcasting CPMs.
- **Penetration (p)** — fraction of vehicles that are CAVs; the primary independent variable (0.0 all-HDV, 1.0 all-CAV).

2. Where we were: Sprints 1–4

Sprint	Delivered	Key outcome
1 RSU infra	Deterministic intersection discovery; single-camera RSU.	Programmatic RSUs on Town03/04/05/10.
2 Detector	YOLO26 fine-tuned on Town03+04+05 (4 classes).	Held-out Town10: mAP@50 0.53, P 0.85, R 0.50.
3 V2X stack	ETSI CPM encoder, Coll-Perales latency, Thandavarayan packet-delivery, edge-compute budget, in-process broker.	Literature-grounded channel; 112 tests.
4 CARLA + ablation	Camera projection, CARLA RSU/CAV, the CAV late-fusion core (time-aware fusion + 3-rule hysteresis + TTC ladder), NHTSA driver profiles, ablation runner.	Penetration sweep + cross-map/weather/VRU extension.

Issues Sprint 5 targets: the Sprint 4 ablation counted collision contact-frames (inflated counts), made every vehicle hostile, spawned some vehicles overlapping, and had no vehicle-to-vehicle link.

3. What Sprint 5 changes

#	Aim	Before	After
1	Collision counter	Contact-frames; stuck cars kept incrementing (~650).	One incident per pair + crashed cars immobilised . Trustworthy counts, no actor-destroy crash.
2	Add V2V	Only RSU -> CAV (V2I).	CAVs broadcast perceived objects (CPM) + pose + hard-brake intent (CAM/DENM); receivers fuse + pre-brake.
3	Spawn overlap	Forced spawn on raw points; overlaps.	try_spawn_actor + minimum-separation filter.
4	Hostility scope	Hostile mix on all vehicles.	HDVs hostile, avoidance OFF; CAVs careful AI_REALISTIC (~1% errors), avoidance ON.
5	New ablation	Single-phase sweep with the issues above.	Two-phase V2V-isolation study (Section 5).

One new module (v2v . py), four extended, runner reworked. **Tests 215 -> 241, all passing.**

4. Methodology

Two-phase isolation. Each cell is run once with the full V2V system and once V2I-only (CAVs keep RSU messages + their own sensor but no CAV-to-CAV link). Differences are attributable to V2V alone.

- **Primary map:** Town05 (in-distribution) at **p = 0.0 / 0.5 / 0.9 / 1.0**, 3 weather presets, 5 seeds — full V2V on/off.
- **Held-out map:** Town10HD_Opt, partial (see Section 6).
- **Population:** 60 vehicles + 60 pedestrians; 3 RSUs; 120 s/cell at max fidelity (20 Hz physics, 10 Hz broadcast).
- **Metrics:** collisions as distinct incidents (total and CAV-involved), braking actions, phantom-brake suppression, cooperative brakes.

Comparability caveat

Sprint 4 counted contact-frames; Sprint 5 counts incidents, so the *absolute* collision numbers are not comparable across sprints. Valid comparisons are **trends** (shape vs penetration) and the **V2V-on vs V2V-off** contrast within Sprint 5.

5. Results

5.1 Town05 — full penetration curve, V2V on/off (mean over 3 weather x 5 seeds; collisions total / CAV-involved).

Penetration	V2I-only coll.	+V2V coll.	V2I hard-brakes	+V2V hard-brakes
0.0 (all HDV)	27.5 / 0.0	27.5 / 0.0	0	0
0.5 (mixed)	14.7 / 7.3	15.5 / 7.3	2,782	13,892
0.9	19.4 / 19.1*	5.6 / 5.1	6,047	58,455
1.0 (all CAV)	2.0 / 2.0	3.2 / 3.2	6,543	45,651

* p=0.9 V2I-only n=14 (one seed missing) and is non-monotonic vs p=0.5 — treat that point as high-variance (see Finding 2).

5.2 Town10HD_Opt — held-out cross-map (partial) (ClearNoon+ClearSunset where both phases exist).

Penetration	V2I-only (tot/cav)	+V2V (tot/cav)	n	note
0.0	41.3 / 0.0	42.0 / 0.0	10	baseline (no CAVs)
0.5	32.2 / 20.8	26.3 / 15.0	9-10	V2V -18% / -28%
1.0	—	8.5 / 8.5	4	V2V only, partial

Findings

Finding 1 — penetration is the dominant safety lever, and it generalises. Replacing hostile HDVs with careful CAVs collapses collisions monotonically. The +V2V Town05 curve is clean: **27.5 -> 15.5 -> 5.6 -> 3.2** across p = 0 / 0.5 / 0.9 / 1.0; the held-out Town10 +V2V curve follows the same shape (42 -> 26 -> 8.5). This role-split safety effect is the headline result and is not a single-map artefact.

Finding 2 — V2V's marginal value depends on scenario difficulty. On easy Town05 at p=0.5 the RSU coverage already catches most conflicts, so V2V is roughly neutral on collisions (14.7 -> 15.5). On the harder held-out Town10 at p=0.5, V2V cuts collisions **-18%** and CAV-involved **-28%** (32.2 -> 26.3). The Town05 p=0.9 point shows a large apparent V2V benefit (19.4 -> 5.6), but the V2I-only baseline there is anomalously high versus p=0.5 and rests on n=14, so we read it as **suggestive, not conclusive**: the +V2V value (5.6) fits the monotone trend, while the V2I spike is likely variance (or a real 'a few hostiles disrupt an un-coordinated CAV fleet' effect) that needs more seeds to separate.

Finding 3 — reactive cooperative braking has a cost. V2V multiplies hard-braking 3-7x, and at full penetration on Town05 it is slightly worse than V2I-only (2.0 -> 3.2): a saturation / string-instability effect of independently-reactive braking, pointing to **coordinated** longitudinal control as the next step.

6. Coverage, and why the full Town10 / Town01 tests are absent

This section is deliberately explicit, because the absence is a performance limit, not an oversight.

6.1 What exists.

- **Town05:** complete — V2V on/off at all four penetrations, 3 weather x 5 seeds.
- **Town10HD_Opt (held-out):** partial — V2V on/off at p0 and p0.5 (n~10); p1.0 only V2V at n=4.
- **Town01:** not run.

6.2 The measured reason. A profiling run (a --profile mode added to the runner) breaks the Town10 p=1.0 cell down by loop section:

Section	V2V on (ms/tick)	V2I-only (ms/tick)
world tick (physics+render)	12.7	14.8
RSU YOLO (GPU)	23.0	19.7
CAV decide (fuse + decode)	207.1	27.5
V2V publish	5.9	n/a
TOTAL	248.9	62.3

The bottleneck is not the GPU (idle at ~8%), not YOLO, and not the V2V *broadcast* (6 ms). It is the **V2V receive path** inside each CAV's per-tick update: at full penetration the 60 CAVs each drain, JSON-decode and fuse roughly 59 incoming messages every period, and the broker's delivery scan is $O(\text{in-flight})$ *per* CAV. As the in-flight queue grows during a run, the per-tick cost grows too — **super-linear** — so a 120 s cell on the denser Town10 map exceeds even a 25-minute per-cell timeout. The work is single-threaded (Python GIL + deterministic synchronous lockstep), so the fast CPU cores and the GPU sit idle while one core does the fusion.

Why not just lower fidelity for Town10?

A reduced tier (5 Hz broadcast + 0.1 s timestep, ~4x less work) does complete Town10 p=1.0 cells. But mixing a reduced-fidelity Town10 with the max-fidelity Town05 study would break the apples-to-apples comparison. We chose to **exclude the full Town10 sweep rather than report mismatched-fidelity numbers**. The held-out Town10 points we do report (p0/p0.5, and p1.0 V2V n=4) are all at the study's full fidelity.

6.3 The fix (future work). The cost is well-localised: index the broker's in-flight queue by receiver so each CAV's delivery drain is $O(\text{its own messages})$ instead of $O(\text{all in-flight})$, and decode each payload once instead of once per receiver. This removes the super-linear term and should make full-fidelity Town10 / Town01 feasible. It was scoped out of Sprint 5 to avoid a broker rewrite late in the cycle.

7. Conclusions and future work

- **Realism fixes worked:** collision counts are trustworthy, roles are physically meaningful, spawns are clean (Sprint 5 aims 1, 3, 4).
- **Safety scales with CAV penetration** and generalises to a held-out map — the clearest, most robust result.
- **V2V helps where the scenario is hard enough to need it**, at a braking-churn cost — motivating coordinated (non-reactive) control.
- **Cooperative-perception fusion cost is the scaling wall:** $O(N^2)$ single-threaded message processing, not the GPU. A receiver-indexed broker is the unlock for full cross-map coverage.

Open items: the broker receiver-index optimisation; completing Town10 and adding Town01 once it lands; more seeds to settle the p=0.9 point; per-detection confidence weighting for lower detector quality on out-of-distribution maps.

8. Appendix — data inventory

Directory (out/)	Contents	Cells
ablation_sprint5_full/v2v	Town05 (full) + Town10 (partial) V2V-on	69
ablation_sprint5_full/noV2V	Town05 (full) V2I-only	45
ablation_sprint5_full/noV2V_t10	Town10 V2I-only p0/p0.5	19
ablation_sprint5_p09/v2v	p=0.9 Town05 V2V-on	15
ablation_sprint5_p09/noV2V	p=0.9 Town05 V2I-only	14

Each directory holds `summary.csv`, `summary_by_cell.csv` and `figures/`. Detailed engineering change-log and the reduced-tier mitigation are in `SPRINT5_RESULTS.md`; profiling breakdowns in `out/speedtest/profile_t10_p100*.json`.
Test suite: 241 passing.